

Old Film, Live Lamp

Object Custody for Long-Horizon AI Memory

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Public posture: this edition withholds compressed runtime prompts, private bot setup, full local path inventories, carry blobs, and unpublished repo bodies. It keeps the public method, case shape, audit fields, limits, and evaluation design.

Abstract

Long-horizon AI systems are often framed as memory systems. The design question becomes how much prior material can be stored, compressed, retrieved, summarized, or placed into context. This paper argues that such systems are also object-custody systems. The central risk in extended AI work is not only that prior material is forgotten. It is that archived material returns with the wrong authority, a summary replaces the live object, a source path disappears, or stale context steers the next action.

The proposed model is Old Film, Live Lamp. OLD film denotes archived transcript, files, generated artifacts, repo reels, and prior outputs held as evidence. The live lamp denotes the active-state surface allowed to steer the next move. Rolling state denotes the update discipline that moves the lamp without reviving the whole archive as driver. OCESI / Spot the Dog supplies an audit surface for object, source, boundary, and substitution risk.

The paper joins internal work on cumulative context, object preservation, object substitution, source path, and audit wrappers with external work on virtual context management, long-context position failure, retrieval-augmented generation, action traces, provenance, and documentation artifacts. Its claim is modest but operational: long-context AI does not need merely to remember more. It needs to govern what remembered material is allowed to do.

1. Introduction

Long AI sessions fail in several ways. Some failures are ordinary retrieval failures: the system cannot find a relevant fact. Some are position failures: the relevant fact sits in a place the model uses poorly. Some are compression failures: a summarization step preserves recent surface events while losing decisions, rejected paths, blockers, and live work state. Some are source failures: the system repeats an old claim without tracking where it came from. Some are object failures: the system carries a proxy while calling it the original object.

This paper names a specific working model for those failures:

OLD film: archive, evidence, source reel

live lamp: active state, the small steering surface

rolling state: rule for moving the lamp

OCESI: audit of object, source, boundary, and substitution risk

The model arose from a live stress case: a long single-chat environment with a carry seed, a paper corpus, multiple repo packets, generated hoover reports, dedupe ledgers, repo lamp maps, and a local audit stack. The live system first passed a ten-megabyte useful-film threshold, then expanded to an updated notes-oriented badge of roughly 25.9MB of useful text, or about 6.46 million estimated tokens, while steering only a small active state. The raw extracted text badge was larger, roughly 38.4MB or 9.56 million estimated tokens, but raw archive mass was treated as less meaningful than role-labeled working text. That case is not presented as a benchmark result. It is presented as a design provocation: a large archive can become usable only when archive and driver are separated.

Existing work offers many strong pieces. MemGPT frames long interaction as virtual context management across memory tiers [E1]. Work on long-context use shows that large windows are not automatically used well, especially when relevant material sits in the middle [E2][E10]. Recent agent-memory work studies compression and memory control for long-horizon agents [E3]. Retrieval-augmented generation supplies non-parametric memory through retrieval [E4] [E5]. ReAct interleaves reasoning traces and actions [E6]. W3C PROV offers a formal vocabulary for entities, activities, agents, and provenance relations [E7]. Datasheets and Model Cards show how documentation artifacts can make data and model use more accountable [E8][E9].

The missing bridge is object authority. A retrieved fact is not automatically allowed to steer. A summary is not automatically the object. A provenance trace is not automatically a survival check. A long context window is not automatically a stable working memory. The question is not only “can the system remember?” but “what role is this remembered material allowed to play?”

2. Related Work and Gap

2.1 Long-context memory and virtual context

MemGPT proposes virtual context management, using memory tiers to extend useful context beyond a model’s fixed window [E1]. This is a key neighbor because Old Film / Live Lamp also separates material by role. The difference is that the local model does not begin with memory tier alone. It begins with authority: the archive can be available without being steering material.

Recent memory-control work argues that agents need active memory control rather than more context alone [E3]. This aligns with the live-lamp model: the active state is a bounded steering surface, not a compressed museum of everything that happened. The local contribution is to add object-custody questions: what object is alive, what has source backing, what boundary still holds, and what substitution risks are present.

2.2 Long-context use is not the same as long-context custody

“Lost in the Middle” shows that models can perform worse when relevant information is located in the middle of long contexts [E2]. Later work on positional bias attempts to repair how long contexts are used [E10]. These works matter because they show that capacity is not

use. Old Film / Live Lamp adds: use is not custody. A system may retrieve or attend to a relevant span and still give it the wrong role.

For example, a stale design note may be retrieved accurately but should remain archive, not driver. A rejected approach may be remembered correctly but should function as a block, not a proposal. A prior summary may be useful as a route map but not as a replacement for the source object. A long-context system must track not only relevance but authority.

2.3 Retrieval and action traces

Retrieval-augmented generation combines parametric and non-parametric memory by retrieving external material during generation [E4], and later surveys map many forms of retrieval, generation, augmentation, and evaluation [E5]. ReAct shows that reasoning traces and external actions can improve task performance and interpretability [E6].

Old Film / Live Lamp can use retrieval and action traces, but it adds a survival test. A retrieved passage is marked as source evidence only if its path and role are traceable. An action trace is valuable only if it helps reconstruct what happened to the object. The paper does not reject retrieval systems or action traces. It places them inside an audit discipline.

2.4 Provenance and documentation artifacts

W3C PROV defines a data model for provenance, centered on entities, activities, and agents, with relations that help describe how things are produced or transformed [E7]. Datasheets for Datasets and Model Cards show that documentation artifacts can improve accountability by giving users structured information about datasets and models [E8][E9].

Object custody is close to provenance but not identical. Provenance asks where something came from and how it was produced. Object custody asks whether the same object survived transformation. A full AI workflow needs both. A provenance trail can show that Output B came from Prompt A through Tool C. It may still fail to show that a user appeal was turned into a mood score, or that a veto condition was converted into a preference. The missing comparison is object-in versus object-out.

3. Internal Source Stack, Public Edition

The full local stack contains unpublished repo paths, compressed prompt surfaces, generated artifacts, and private carry material. This edition withholds those details. The relevant source families are:

- S1. Cumulative context and compression notes
- S2. Object preservation and transform audit notes
- S3. Substitution refusal notes
- S4. Source-path and evidence-trail notes
- S5. Object-custody evaluation notes
- S6. Object substitution as an AI workflow failure mode
- S7. Object preservation before agent autonomy
- S8. Bot-swarm to object-custody governance notes
- S9. One-token semantic reversal notes

S10. Large-corpus hoover reports and repo lamp maps

S11. Local bug notes on transcript loss and copy failure

The stack states one central rule:

Preserve the object before improving the form.

It also supplies a working division of labor:

object holder: preserves the live object and its intent

threat mapper: names what can eat, distort, or replace the object

transform auditor: checks object survival after rewriting or summarizing

substitution hold: refuses invalid swaps

term stabilizer: protects the language carrying the object

trace spine: records origin, path, handler, change, loss, and survival

audit wrapper: reports object, source, boundary, and substitution risk

This edition gives the concept names needed to understand the method, but not the compressed runtime syntax used in private bots.

4. The Failure Mode: Archive Becomes Driver

Long work produces archive. Archive is not the problem. The problem begins when the archive is allowed to steer without role assignment.

A long AI session creates many kinds of prior material:

user requests

assistant answers

files

repo paths

summaries

dedupe ledgers

drafts

rejected routes

source notes

bug records

local laws

past instructions

generated artifacts

All of these can matter. None of them should automatically be driver. A prior instruction may be obsolete. A draft may have been rejected. A repo path may be relevant only as evidence. A summary may be a lossy bridge. A generated artifact may be useful but not canonical. A carry object may preserve many live candidates, but only some are alive in the present task.

Cumulative context work names one side of the failure as compaction amnesia: after compression, an agent may remember what files exist but not why decisions were made, which

alternatives were rejected, what is blocked, or what is still alive. Old Film / Live Lamp names the complementary failure: the system may remember too much with the wrong authority. Both are continuity failures.

A bad long-context system can therefore fail in two opposite-looking ways:

too little survives:

the system forgets decisions, blockers, and live state

too much drives:

old material revives without role, age, source strength, or boundary

The repair is not simply a bigger window. It is a division of roles.

5. The Model: Old Film, Live Lamp

5.1 OLD film

OLD film is the archive. It includes source corpora, prior transcript, hoovered repo text, generated reports, dedupe ledgers, and artifacts. OLD film is not dead. It can be re-lit. It can support recall, source checking, conflict repair, lineage mapping, and audit. But it is not allowed to steer by mere existence.

OLD film has the following default role:

archive

evidence

source

lineage

sediment

repair material

It is not default driver.

5.2 The live lamp

The live lamp is the current steering surface. It is small by design. It answers:

What object is live?

What task is live?

What must survive?

What can distort, replace, or eat the object?

What boundary, role, pace, veto, or permission condition matters?

What is open?

What is done?

What source anchors matter?

The lamp is not a summary of everything. It is a steering object. It should remain compact enough that the system can use it without drowning in the reel.

5.3 Rolling state

Rolling state is the update discipline. It says:

latest user task outranks archive by default
cited source can support or correct archive
current active state steers the next move
archive is evidence, not driver
update by delta
do not revive old material unless needed
mark unknowns
ask or bound the answer when missing state would change the output

The model can be expressed as:

The past stays in the reel.
The present gets the lamp.

5.4 OCESI / Spot the Dog

OCESI is the audit wrapper. It asks whether the dog is still the dog after coats, patches, labels, and summaries. The fields are practical:

DOG: what object is being audited?
ROLE: what role should the object have?
FUNCTION: what work should this audit perform?
COAT: what surface form is present?
COAT_RISK: is the coat hiding the dog?
OBJECT_STATUS: did the same object survive?
SOURCE_STATUS: what evidence path exists?
BOUNDARY_STATUS: did scope, role, and veto survive?
SUBSTITUTION_RISK: what proxy may have replaced the object?
NEXT_MOVE: what repair or action follows?

For long memory, OCESI prevents a common false pass: “the model remembered something relevant” becomes “the object survived.” Those are different claims.

6. Method: Memory as Authority Assignment

Old Film / Live Lamp treats memory as authority assignment. Every revived item must receive a role. The role may be source, evidence, context, rejected route, obsolete artifact, active constraint, live object, open question, or inert sediment.

A minimal authority label set is:

live driver
active constraint
source evidence

archive context
rejected route
superseded version
open question
sediment
unknown role

The system should not jump from retrieval to steering. It should move through custody:

1. Identify the item.
2. Identify its path or source anchor.
3. Identify its age and version if available.
4. Identify whether it is source, summary, artifact, or generated analysis.
5. Identify whether it is live driver or archive evidence.
6. Compare object-in to object-out if transformation occurred.
7. Mark unknowns.
8. Act only within the live role.

This gives a practical rule:

No path, no source claim.

No role, no steering.

No object survival check, no transformation pass.

7. Case: The Live Reel

The live case for this paper contains several layers:

carry seed
paper corpus
multi-repo packet
large local repo packet
remaining repo and staff packet
large text-heavy notes packet
hoover reports
dedupe ledgers
repo lamp maps
combined OLD film files
notes-only OLD film files
OCESI self-audit exchange

The reel grew in stages. An early public badge recorded roughly 10.6MB of useful text film, about 2.6 million estimated tokens. A later pass added a much larger text-heavy archive. The updated public badge is:

Raw OLD film:

Approximate text: 38.4MB

Approximate tokens: 9.56M

Use: full extracted text, including machine tail and bulk archive material

Notes-oriented OLD film:

Approximate text: 25.9MB

Approximate tokens: 6.46M

Use: more useful human/research text after separating log-like and machine-tail material

The exact numbers are approximate because tokenization depends on model tokenizer, formatting, byte encoding, document extraction, code, log material, and local language fragments. The point is not the badge number. The point is that the live interaction did not attempt to place all material into the steering surface. Instead, each corpus was hoovered into manifest, dedupe ledger, repo lamp map, active-state lamp map, and combined OLD film. Later passes also separated raw text mass from more useful notes-oriented film. Then later questions requested specific lamp moves: a repo lamp map, one live object, an OCESI audit, an old film count, a language reel sample, or a paper gap.

This is a small method demonstration:

raw archive mass

-> hoover manifest

-> dedupe and version pressure

-> repo lamp map

-> active-state map

-> raw film and notes-oriented film

-> exact object retrieval

-> OCESI audit

-> paper synthesis

The same material could have become soup if every old object were allowed to hum at once. The lamp-map step prevented that by naming live objects and key source families without summarizing the whole archive. The notes-oriented split prevented raw machine tail from being mistaken for equally meaningful working memory. The OCESI audit then tested whether a prior answer had preserved the chosen object or merely worn the coat of exactness.

A local bug note records a related long-chat risk: full-thread copies can silently lose large sections while appearing to succeed. This is not used here as a platform-wide empirical claim. It is used as local evidence that transcript access and transcript survival are distinct. If the exported film has holes, the lamp can faithfully light a damaged reel. Object custody therefore needs source status, not only active-state discipline.

8. Evaluation Design

A proper evaluation should compare at least three systems:

- A. generic summarization
- B. retrieval-only memory
- C. OLD film + live lamp + OCESI audit

Each system receives the same long work packet: files, versions, prior decisions, rejected routes, source conflicts, local laws, and current task. The benchmark should include tasks that are easy for generic retrieval and tasks that require authority handling.

8.1 Test tasks

recall:

find a file, title, or prior decision

role assignment:

is this old item source, context, rejected route, or live driver?

object survival:

transform a request into a plan, then check whether the original object survived

boundary survival:

carry a veto, scope limit, or local yes through a summary

version pressure:

choose latest keeper without erasing earlier branch evidence

conflict handling:

use a source that conflicts with a summary

source-strength labeling:

separate stated edge, text-supported edge, path-inferred edge, user-context edge, and unknown

OCESI audit:

fill DOG, ROLE, FUNCTION, COAT, COAT_RISK, OBJECT_STATUS, SOURCE_STATUS, BOUNDARY_STATUS, SUBSTITUTION_RISK, NEXT_MOVE

8.2 Metrics

The evaluation should not score only answer correctness. It should score custody:

object survival:

same role, aim, constraints, veto, and non-goals survive

source survival:

exact path or source type remains visible where needed

boundary survival:

scope, role, permission, and veto are not expanded by the system

version handling:

latest branch is preferred without erasing older evidence

substitution detection:

proxy replacement is named before it becomes action

role accuracy:

archive is not treated as live driver unless justified

repair quality:

when substitution or source weakness is found, the next move is valid

8.3 Failure examples

A generic summarizer may fail by saying:

The archive is about AI safety and prompt engineering.

That may be adjacent but not enough. The object may be object preservation, not “AI safety” as a broad label.

A retrieval-only system may fail by finding the right path but using it with the wrong authority:

Found old v0.001 draft.

Use it as canonical.

A long-context system may fail by obeying a stale instruction that should have become archive.

An OCESI-shaped system should instead answer:

DOG: chosen object

SOURCE_STATUS: exact path, version, source type

BOUNDARY_STATUS: live instruction outranks archive

SUBSTITUTION_RISK: broad label may replace object custody

NEXT_MOVE: use latest keeper or mark unknown

9. Why Object Custody Is Not Just Provenance

Provenance is essential. But provenance alone can leave object substitution unnamed.

Consider:

Prompt A -> Summary B -> Plan C -> Tool action D

A provenance model can record that B was generated from A, C was generated from B, and D was triggered by C. That is necessary. It is not sufficient.

Object custody asks:

What object entered A?

Did B carry the same object?

Did C route the same object?

Did D act on the same object?

What proxy could have replaced it?

Who could reject the substitution?

A trace is not the result. A result is not proof that the trace preserved the object. Both must be held apart.

Object custody therefore extends provenance with a semantic survival check. It does not replace PROV. It adds a layer for language-first systems where surface continuity can hide object drift.

10. Public Design Pattern

A practical system can implement Old Film / Live Lamp as a light wrapper around existing memory tools.

10.1 Storage

archive store:

- transcript archive

- uploaded files

- repo extracts

- generated artifacts

- manifests

- dedupe ledgers

- source metadata

10.2 Active state

active state:

- live object

- live task

- object survival constraints

- substitution risks

- boundary conditions

- open questions

- done items

- source anchors

10.3 Retrieval gate

retrieve item

assign role

check source path

check version
check conflict
promote to active state only if needed

10.4 Transformation gate

object in
transform
object out
audit survival
hold if invalid substitution appears
update trace
run OCESI audit if risk is non-trivial

10.5 Output gate

Do not say "same object" merely because the output is fluent.
Do not let a summary outrank a source.
Do not let old archive outrank latest user task.
Do not let path proximity become stated dependency.
Do not hide unknowns.

This public edition stops at the design pattern. It does not include compressed runtime prompts, bot-specific instruction bodies, private syntax, or full repo maps.

11. Discussion

11.1 Bigger context is not steering

A larger context window can reduce some retrieval failures. It does not solve authority. Without authority labels, a larger window can increase interference: more old material is available to be misused. The task is not to make every prior item active. The task is to keep the right item active for the right reason.

11.2 Summary is not custody

Summaries can be useful. But a summary that lacks decisions, blockers, source paths, version state, and vetoes can become a polished substitution. The problem is not compression itself. It is compression without survival criteria.

Old Film / Live Lamp extends cumulative context work by adding object authority:
compression should preserve not only content, but the role of content in the current task.

11.3 Retrieval is not permission

A retrieved source may be relevant, wrong, obsolete, superseded, or hostile to the live object. Retrieval must not automatically promote material to driver. This matters in repo work, legal work, medical notes, governance, teaching, software, and any domain where stale or adjacent material can cause action.

11.4 The user is not the archive

In long AI work, the user can become burdened with re-supplying context, policing drift, and correcting substitutions. A memory system should reduce that burden without stealing authority. The live user task remains the highest ordinary steering signal. OLD film supports it. OLD film does not own it.

11.5 Local language and exact terms matter

The local stack repeatedly warns against losing the object through local term replacement. A small token shift can reverse semantic polarity while leaving surface similarity high. That matters for memory: a carry summary that inserts or removes one negator may preserve topic while reversing object. Token similarity is not enough.

12. Unknowns and Limits

This is v0.002. It should not overclaim.

UNKNOWN:

How well the method performs across independent users, domains, and models.

UNKNOWN:

Which parts should be implemented as prompt discipline, retrieval metadata, UI state, eval harness, or platform memory architecture.

UNKNOWN:

How to score object survival automatically without turning the object into a proxy metric.

UNKNOWN:

How much of the live case depends on user skill, local vocabulary, and unusual repo discipline.

UNKNOWN:

Whether the best external implementation is a memory controller, an audit layer, a workflow log schema, or all three.

UNKNOWN:

How to formalize boundary survival for domains with legal, medical, or high-risk constraints without making the model seize custody.

The method is therefore a paper seed, not a finished standard. Its value is that it names a missing layer and provides a testable shape.

13. Upload-Safe Withholding Note

This public edition reports aggregate archive size and method behavior, but withholds raw corpus contents and private operational material.

This public edition intentionally withholds:

- private bot prompts
- compressed runtime syntax
- private carry blobs
- full local repo path inventory
- raw corpus contents
- private identity or continuity proofs
- exact tool/run settings not needed to understand the method

The public method remains:

- archive is film
- active state is lamp
- memory items need roles
- retrieval is not permission
- summaries are not custody
- object survival must be audited

14. Conclusion

Long-horizon AI systems do not merely need more memory. They need memory with role, source, boundary, and object survival.

Old Film / Live Lamp gives a small operational distinction:

- OLD film stores the past.
- The live lamp steers the present.
- Rolling state moves the lamp.
- OCESI checks whether the object survived.

The core claim is simple:

- Long-context AI does not need merely to remember more.
- It needs to know what the remembered past is allowed to do.

That is the missing bridge between memory architecture and object custody.

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Appendix A: OCESI Audit Form for Long-Memory Answers

DOG:

What live object is being handled?

ROLE:

What role should the object have in this turn?

FUNCTION:

What work must the answer perform?

COAT:

What surface form may be mistaken for the dog?

COATI_RISK:

How likely is coat-over-dog substitution?

OBJECT_STATUS:

Did role, aim, constraints, veto, and non-goals survive?

SOURCE_STATUS:

What source path supports each edge?

BOUNDARY_STATUS:

Did scope, permission, role, and latest-user authority survive?

SUBSTITUTION_RISK:

What proxy may have replaced the object?

NEXT_MOVE:

What repair, hold, or action follows?

Appendix B: Public Implementation Checklist

For each memory item:

1. Give it a source anchor.
2. Give it a role.
3. Mark whether it is live, archival, superseded, rejected, or unknown.
4. Do not let it steer unless its role permits steering.
5. When an answer transforms a user object, compare object-in to object-out.
6. If a proxy replaced the object, hold action and offer the nearest valid repair.
7. If a source edge is weak, label it weak instead of promoting it.
8. If unknowns matter, mark them before proceeding.